

Quiz Survival Guide

Matrix Calculus, Least Squares, Circle/Sphere/Ellipse Fitting

Built strictly from the two supplied lecture PDFs

What this PDF does: ranks likely quiz targets by study priority, gives deliberately tricky rephrasings, lists accepted answer forms first, shows common distractors, explains why the answer works, and gives a short memory hook for every item.

Prepared for last-minute review

Priority labels are an exam-prep inference from emphasis/repetition in the notes, not a claim about the actual quiz.

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How to use this tonight

1. First memorize the one-page formula map below.
2. Then do Questions 1–35 in order: the highest-yield material is intentionally first.
3. For each question, cover the answers and try every “tricky formulation” aloud.
4. Finish with Questions 39–40 so source typos and mixed-method recognition cannot catch you off guard.

Source-faithfulness rule used here. The guide follows the supplied PDFs. When a printed line is internally inconsistent with the equation immediately around it, it is explicitly flagged rather than silently rewritten. Derived material is labeled “derived” rather than presented as if it were printed verbatim in the notes.

Ultra-fast formula map

Topic	Formula / signature
Vector gradient	$\frac{\partial J}{\partial \mathbf{x}} = [\partial J / \partial x_1, \dots, \partial J / \partial x_N]^T$
$\mathbf{x}^T \mathbf{x}$	$\frac{\partial}{\partial \mathbf{x}} (\mathbf{x}^T \mathbf{x}) = 2\mathbf{x}$
$\mathbf{x}^T \mathbf{a}$	$\frac{\partial}{\partial \mathbf{x}} (\mathbf{x}^T \mathbf{a}) = \mathbf{a}$
Quadratic form	$\frac{\partial}{\partial \mathbf{x}} (\mathbf{x}^T A \mathbf{x}) = (A^T + A)\mathbf{x}$; if symmetric B : $2B\mathbf{x}$
Least squares	$\min \ A\mathbf{x} - \mathbf{b}\ _2^2$, normal equation $A^T A \mathbf{x} = A^T \mathbf{b}$
LS solution	$\mathbf{x} = (A^T A)^{-1} A^T \mathbf{b}$ (source formula)
Line	$y = mx + b$, row $[x_i, 1]$
Plane	$z = ax + by + c$, row $[x_i, y_i, 1]$
Circle	$(x - x_0)^2 + (y - y_0)^2 = R^2$, row $[1, -2x_i, -2y_i]$
Sphere	Circle row plus $-2z_i$; grouped parameter also gains z_0
Ellipse	$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$, condition $B^2 - 4AC < 0$
Ellipse row	$[x_i^2, x_i y_i, y_i^2, x_i, y_i, 1]$, $M\mathbf{z} = 0$
Ellipse constraint	$B^2 - 4AC = -1 \iff \mathbf{z}^T K \mathbf{z} = -1$
Ellipse solve	$M^T M \mathbf{z} = -\lambda K \mathbf{z}$ (generalized eigenvalue problem)
Circle normal	$\nabla J = [2(x - x_0), 2(y - y_0)]^T$
Ellipse normal	$\nabla J = [2Ax + By + D, Bx + 2Cy + E]^T$ (see source-typo alert)

Ranked question bank

Q1. Least-squares objective for an over-determined linear system – Priority: VERY HIGH

Source anchor: Matrix notes, Sec. 3

Possible tricky formulations

- How is an over-determined system $A\mathbf{x} = \mathbf{b}$ solved in the least-squares sense?
- Which optimization problem gives the standard estimate when $m > n$?
- Choose the cost function minimized when the linear equations cannot all be satisfied exactly.
- What does “least squares minimize: $\|A\mathbf{x} - \mathbf{b}\|_2$, its square, or something else?

Correct / accepted:

- $\mathbf{x} = \arg \min_{\mathbf{x}} \|A\mathbf{x} - \mathbf{b}\|_2^2$ (standard source form).
- Equivalent minimizer: $\arg \min_{\mathbf{x}} \|A\mathbf{x} - \mathbf{b}\|_2$; squaring does not change which nonnegative residual norm is smallest.

Likely traps / distractors:

- $\arg \min \|A\mathbf{x} + \mathbf{b}\|_2^2$ (wrong sign).
- $\arg \max \|A\mathbf{x} - \mathbf{b}\|_2^2$ (opposite goal).
- $\arg \min \|\mathbf{x}\|_2^2$ (different problem).

Why it is correct. The notes define the residual as $A\mathbf{x} - \mathbf{b}$ and minimize its squared Euclidean norm when the system has more equations than unknowns.

Memory / understanding hook. Least squares means: make the residual vector as short as possible; “square is there for easy differentiation.

Q2. Normal equation – Priority: VERY HIGH

Source anchor: Matrix notes, Sec. 3

Possible tricky formulations

- What equation is obtained by setting the gradient of the least-squares cost to zero?
- Which of the following is called the normal equation?
- After differentiating $J = \|A\mathbf{x} - \mathbf{b}\|_2^2$, what linear system must $\mathbf{x}$ satisfy?

Correct / accepted:

- $A^T A \mathbf{x} = A^T \mathbf{b}$.

Likely traps / distractors:

- $AA^T \mathbf{x} = \mathbf{b}$.
- $A^T \mathbf{x} = A\mathbf{b}$.
- $A\mathbf{x} = A^T \mathbf{b}$.

Why it is correct. The gradient in the notes is $2A^T A \mathbf{x} - 2A^T \mathbf{b}$; setting it to zero gives the normal equation.

Memory / understanding hook. Transpose the original system from the left: both sides get A^T .

Q3. Closed-form least-squares solution – Priority: VERY HIGH

Source anchor: Matrix notes, Sec. 3

Possible tricky formulations

- Solve the normal equation for $\mathbf{x}$.
- Which formula is the standard least-squares solution shown in the notes?
- What expression acts as the left pseudo-inverse in the notes?

Correct / accepted:

- $\mathbf{x} = (A^T A)^{-1} A^T \mathbf{b}$.
- $\mathbf{x} = A^+ \mathbf{b}$ if the pseudo-inverse notation A^+ is used (same least-squares idea; the notes explicitly mention pseudo-inverses in fitting problems).

Likely traps / distractors:

- $(AA^T)^{-1} A \mathbf{b}$.
- $A^{-1} \mathbf{b}$ for a generic non-square A .
- $A^T (AA^T)^{-1} \mathbf{b}$ as the formula printed in these notes.

Why it is correct. The notes solve $A^T A \mathbf{x} = A^T \mathbf{b}$ by multiplying by $(A^T A)^{-1}$.

Memory / understanding hook. Normal equation first, then “invert the normal matrix $A^T A$ ”.

Q4. Derivative of a quadratic vector norm – Priority: VERY HIGH

Source anchor: Matrix notes, Sec. 2.1.1

Possible tricky formulations

- Compute $\partial(\mathbf{x}^T \mathbf{x}) / \partial \mathbf{x}$.
- What is the gradient of the squared L_2 norm?
- If $J = \sum_i x_i^2$, what is $\nabla_{\mathbf{x}} J$?

Correct / accepted:

- $\frac{\partial(\mathbf{x}^T \mathbf{x})}{\partial \mathbf{x}} = 2\mathbf{x}$.
- $\nabla_{\mathbf{x}} \|\mathbf{x}\|_2^2 = 2\mathbf{x}$.

Likely traps / distractors:

- $\mathbf{x}$.
- $2\mathbf{x}^2$ componentwise.
- 1.

Why it is correct. Each component derivative is $\partial x_i^2 / \partial x_i = 2x_i$, so the gradient stacks into $2\mathbf{x}$.

Memory / understanding hook. Scalar $x^2 \rightarrow 2x$; vector $\mathbf{x}^T \mathbf{x} \rightarrow 2\mathbf{x}$.

Q5. Derivative of a scalar product – Priority: VERY HIGH

Source anchor: Matrix notes, Sec. 2.1.2

Possible tricky formulations

- Differentiate $\mathbf{x}^T \mathbf{a}$ with respect to $\mathbf{x}$.
- What is $\partial(\mathbf{a}^T \mathbf{x}) / \partial \mathbf{x}$?
- Which vector is the gradient of $J = \sum_i a_i x_i$?

Correct / accepted:

- $\frac{\partial(\mathbf{x}^T \mathbf{a})}{\partial \mathbf{x}} = \mathbf{a}$.
- $\frac{\partial(\mathbf{a}^T \mathbf{x})}{\partial \mathbf{x}} = \mathbf{a}$.

Likely traps / distractors:

- $\mathbf{x}$.
- $2\mathbf{a}$.
- $\mathbf{a}^T$ as a row when the notes define the derivative as a column vector.

Why it is correct. The coefficient multiplying each unknown x_i is a_i , so each derivative component equals a_i .

Memory / understanding hook. The gradient of a linear form returns its coefficient vector.

Q6. Derivative of $\mathbf{x}^T A \mathbf{x}$ – Priority: VERY HIGH*Source anchor: Matrix notes, Sec. 2.1.3***Possible tricky formulations**

- Find $\partial(\mathbf{x}^T A \mathbf{x})/\partial \mathbf{x}$ for a general square matrix A .
- Which formula is correct when A is not assumed symmetric?
- Why do both A and A^T appear in the derivative of a quadratic form?

Correct / accepted:

$$\frac{\partial(\mathbf{x}^T A \mathbf{x})}{\partial \mathbf{x}} = (A^T + A)\mathbf{x}.$$

Likely traps / distractors:

- $2A\mathbf{x}$ for arbitrary A (only the symmetric special case).
- $A\mathbf{x}$.
- $(A^T - A)\mathbf{x}$.

Why it is correct. Each x_k appears once through row k and once through column k , producing contributions from A and A^T .

Memory / understanding hook. General quadratic form: symmetrize the matrix in the gradient, $A + A^T$.

Q7. Symmetric quadratic-form special case – Priority: VERY HIGH*Source anchor: Matrix notes, Sec. 2.1.3***Possible tricky formulations**

- If $B^T = B$, simplify $\partial(\mathbf{x}^T B \mathbf{x})/\partial \mathbf{x}$.
- When can $(A^T + A)\mathbf{x}$ become $2A\mathbf{x}$?
- What does symmetry buy you in the quadratic-form derivative?

Correct / accepted:

- $2B\mathbf{x}$ when $B^T = B$.

Likely traps / distractors:

- $B\mathbf{x}$.
- $2B^T \mathbf{x}$ presented as a different answer (it is equal only because B is symmetric).
- 0 .

Why it is correct. Substitute $B^T = B$ into $(B^T + B)\mathbf{x}$.

Memory / understanding hook. Symmetric means the two derivative contributions are identical, so they add to two copies.

Q8. Definition of the L_2 norm – Priority: HIGH*Source anchor: Matrix notes, Sec. 3***Possible tricky formulations**

- How is the Euclidean norm of a vector $\mathbf{v}$ written using a transpose?
- Which norm is used in the least-squares derivation?
- Complete: $\|\mathbf{v}\|_2 = ?$

Correct / accepted:

$$\|\mathbf{v}\|_2 = \sqrt{\mathbf{v}^T \mathbf{v}}.$$

Likely traps / distractors:

- $\mathbf{v}^T \mathbf{v}$ (this is the squared norm).
- $\sqrt{\mathbf{v} \mathbf{v}^T}$ (matrix/scalar mismatch in general).

- $\sum_i |v_i|$ (the L_1 norm).

Why it is correct. The source explicitly calls $\sqrt{\mathbf{v}^T \mathbf{v}}$ the L_2 norm or Euclidean distance.

Memory / understanding hook. Dot the vector with itself, then take the square root.

Q9. What makes a linear system over-determined? – Priority: HIGH

Source anchor: Matrix notes, Sec. 3

Possible tricky formulations

- For $A \in \mathbb{R}^{m \times n}$, when is the system over-determined?
- What relation between m and n leads to an estimation problem in the notes?
- More equations than unknowns corresponds to which inequality?

Correct / accepted:

- $m > n$: more rows/equations than unknowns.

Likely traps / distractors:

- $m < n$.
- $m = n$.
- $m = 0$.

Why it is correct. The notes state that when a non-square matrix has $m > n$, the problem is an estimation problem and the mismatch should be minimized.

Memory / understanding hook. Rows are equations, columns are unknowns; over-determined means rows $>$ columns.

Q10. Gradient of a scalar function with respect to a vector – Priority: HIGH

Source anchor: Matrix notes, Sec. 2

Possible tricky formulations

- How do the notes define $\partial J / \partial \mathbf{x}$?
- Is the derivative of a scalar with respect to a vector a scalar, row, or column in this convention?
- Write the component form of the gradient.

Correct / accepted:

- $\frac{\partial J}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial J}{\partial x_1} & \cdots & \frac{\partial J}{\partial x_N} \end{bmatrix}^T$.

Likely traps / distractors:

- A single scalar derivative.
- A row vector in the convention shown in these notes.
- $J\mathbf{x}$.

Why it is correct. The notes stack the partial derivatives with respect to every coordinate into a column vector.

Memory / understanding hook. One scalar cost, many coordinates: collect one partial derivative per coordinate into a column.

Q11. Line fitting: model and design matrix – Priority: VERY HIGH

Source anchor: Matrix notes, Sec. 4.1

Possible tricky formulations

- Write the linear system used to fit $y = mx + b$ to N points.
- What is the design matrix for 2D slope-intercept line fitting?
- Which columns correspond to slope and intercept?

Correct / accepted:

- $y = mx + b$.

- $$\begin{bmatrix} x_1 & 1 \\ x_2 & 1 \\ \vdots & \vdots \\ x_N & 1 \end{bmatrix} \begin{bmatrix} m \\ b \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix}.$$

Likely traps / distractors:

- Use rows $[y_i, 1]$ and right side x_i .
- Use $[x_i, y_i]$ with unknowns $[m, b]^T$.
- Omit the column of ones, which would remove the intercept.

Why it is correct. Each point contributes the equation $y_i = mx_i + b$, which is linear in the unknown parameters m and b .

Memory / understanding hook. Slope multiplies x ; intercept multiplies 1.

Q12. Line fitting: least-squares solution – Priority: VERY HIGH

Source anchor: Matrix notes, Sec. 4.1

Possible tricky formulations

- Given the line design matrix X , how are m and b estimated?
- Which normal-equation formula applies to line fitting?
- Express $[m \ b]^T$ using X and $\mathbf{y}$.

Correct / accepted:

- $$\begin{bmatrix} m \\ b \end{bmatrix} = (X^T X)^{-1} X^T \mathbf{y} \text{ with } X = [x_i \ 1].$$

Likely traps / distractors:

- $X^{-1}\mathbf{y}$ for a tall X .
- $XX^T\mathbf{y}$.
- $(X^T X)X^T\mathbf{y}$ without inversion.

Why it is correct. Line fitting is presented as a direct case study of the general least-squares formula.

Memory / understanding hook. Build X from x and ones, then reuse the universal least-squares template.

Q13. Line fitting: expanded normal matrix – Priority: HIGH

Source anchor: Matrix notes, Sec. 4.1

Possible tricky formulations

- What does $X^T X$ become for line fitting?
- Which sums appear in the closed form for m and b ?
- Choose the correct 2×2 normal matrix.

Correct / accepted:

- $$X^T X = \begin{bmatrix} \sum x_i^2 & \sum x_i \\ \sum x_i & N \end{bmatrix}.$$
- $$X^T \mathbf{y} = \begin{bmatrix} \sum x_i y_i \\ \sum y_i \end{bmatrix}.$$

Likely traps / distractors:

- Put $\sum y_i^2$ in $X^T X$.
- Use N^2 in the lower-right entry.
- Use $\sum x_i y_i$ as an off-diagonal entry of $X^T X$.

Why it is correct. Multiplying the two columns x and 1 against each other gives $\sum x_i^2$, $\sum x_i$, and N .

Memory / understanding hook. Normal matrix = all pairwise dot products of the design-matrix columns.

Q14. Plane fitting: model and design matrix – Priority: VERY HIGH*Source anchor: Matrix notes, Sec. 4.2***Possible tricky formulations**

- Write the explicit plane model used in the notes.
- How do N 3D points form a linear system for a, b, c ?
- Which row of the plane design matrix corresponds to point (x_i, y_i, z_i) ?

Correct / accepted:

- $z = ax + by + c$.
- $$\begin{bmatrix} x_1 & y_1 & 1 \\ \vdots & \vdots & \vdots \\ x_N & y_N & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} z_1 \\ \vdots \\ z_N \end{bmatrix}.$$
- A single row is $[x_i \ y_i \ 1]$ with right side z_i .

Likely traps / distractors:

- $z = ax + cy + b$ but then keep parameters in the order $[a, b, c]$ without changing the matrix.
- Use row $[x_i, y_i, z_i]$ and right side 1.
- Omit the ones column.

Why it is correct. Each point supplies $z_i = ax_i + by_i + c$, which is linear in the three plane parameters.**Memory / understanding hook.** Plane fitting is line fitting with one extra coordinate column: x , y , and 1.**Q15. Plane fitting: expanded least-squares solution – Priority: HIGH***Source anchor: Matrix notes, Sec. 4.2***Possible tricky formulations**

- Which sums appear in the 3×3 normal matrix for plane fitting?
- Write $X^T X$ and $X^T \mathbf{z}$ for the plane model.
- How are a, b, c obtained from the data sums?

Correct / accepted:

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} \sum x_i^2 & \sum x_i y_i & \sum x_i \\ \sum x_i y_i & \sum y_i^2 & \sum y_i \\ \sum x_i & \sum y_i & N \end{bmatrix}^{-1} \begin{bmatrix} \sum x_i z_i \\ \sum y_i z_i \\ \sum z_i \end{bmatrix}.$$

Likely traps / distractors:

- Put $\sum z_i^2$ into the design normal matrix.
- Use an asymmetric normal matrix.
- Replace the last diagonal N by 1.

Why it is correct. As with line fitting, $X^T X$ contains dot products of the columns x , y , and 1, while $X^T \mathbf{z}$ correlates each column with the measured z .**Memory / understanding hook.** For $[x, y, 1]$, every entry is a sum of products of those three column types.**Q16. Circle: implicit equation and expansion – Priority: VERY HIGH***Source anchor: Circle/Ellipse notes, Sec. 1.1***Possible tricky formulations**

- What is the implicit equation of a circle with center (x_0, y_0) and radius R ?
- Expand the circle equation into terms linear in the unknown grouped parameters.
- Which terms produce the coefficients $-2x_0$ and $-2y_0$?

Correct / accepted:

- $(x - x_0)^2 + (y - y_0)^2 = R^2$.
- $x^2 - 2x_0x + x_0^2 + y^2 - 2y_0y + y_0^2 - R^2 = 0$.

Likely traps / distractors:

- $(x + x_0)^2 + (y + y_0)^2 = R^2$.
- $x^2 + y^2 = R$ (missing center and squared radius).
- $x^2 - 2x_0 + x_0^2 + \dots$ (missing multiplication by x).

Why it is correct. Expanding the squared coordinate differences gives a form where the data coordinates multiply unknown center parameters linearly.

Memory / understanding hook. Center coordinates enter through the middle terms of $(x - x_0)^2$ and $(y - y_0)^2$.

Q17. Circle fitting as an inhomogeneous linear system – Priority: VERY HIGH

Source anchor: Circle/Ellipse notes, Sec. 1.1

Possible tricky formulations

- What matrix system is used to fit a circle to points (x_i, y_i) ?
- Which unknown vector makes circle fitting linear?
- Choose the correct row and right-hand side for point i .

Correct / accepted:

- $\begin{bmatrix} 1 & -2x_i & -2y_i \end{bmatrix} \begin{bmatrix} x_0^2 + y_0^2 - R^2 \\ x_0 \\ y_0 \end{bmatrix} = -(x_i^2 + y_i^2)$ for every i .
- Stacking all rows gives an inhomogeneous system $C\mathbf{p} = \mathbf{d}$.

Likely traps / distractors:

- Use row $[1, 2x_i, 2y_i]$ without changing the parameter signs.
- Use right side $+(x_i^2 + y_i^2)$.
- Use $[x_0, y_0, R]^T$ directly as a linear parameter vector in this formulation.

Why it is correct. The expanded circle equation is rearranged so all unknown grouped parameters are on the left and known squared coordinates are on the right.

Memory / understanding hook. Circle row: $[1, -2x, -2y]$; right side: negative squared distance from the origin.

Q18. How circle fitting is solved – Priority: VERY HIGH

Source anchor: Circle/Ellipse notes, Sec. 1.1

Possible tricky formulations

- Which numerical tool is suggested for an over-determined circle fit?
- Inverse or pseudo-inverse: which one applies when there are many points?
- Is circle fitting here a homogeneous or inhomogeneous linear problem?

Correct / accepted:

- It is an **inhomogeneous** linear problem.
- Use the inverse when appropriate, or the **pseudo-inverse** in the over-determined case.

Likely traps / distractors:

- Solve a generalized eigenvalue problem (that is the ellipse method in these notes).
- It is homogeneous because the right side is zero.
- No linear algebra is needed.

Why it is correct. The circle system has a nonzero right-hand side and the notes explicitly recommend inverse/pseudo-inverse solution.

Memory / understanding hook. Circle = nonzero right-hand side = least-squares/pseudo-inverse.

Q19. Sphere: 3D analogue of circle fitting – Priority: HIGH*Source anchor: Circle/Ellipse notes, Sec. 1.1.1***Possible tricky formulations**

- How is the circle equation extended to a sphere?
- What is the 3D counterpart of the circle center and radius model?
- Which coordinate gets added when going from circle to sphere fitting?

Correct / accepted:

- Intended 3D model: $(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = R^2$.
- Center vector: $[x_0 \ y_0 \ z_0]^T$.

Likely traps / distractors:

- Use $(z - y_0)^2$ as the intended geometric model (this appears as a source typo on page 1, while the expansion on page 2 uses z_0).
- Drop the z term.
- Use a different radius in each coordinate.

Why it is correct. The sphere is described as the 3D version of the circle, and the following expanded equation clearly uses $-2z_0z + z_0^2$.

Memory / understanding hook. Sphere = circle plus one matching pair: $(z - z_0)^2$.

Q20. Sphere fitting linear system – Priority: HIGH*Source anchor: Circle/Ellipse notes, Sec. 1.1.1***Possible tricky formulations**

- What row is added for sphere fitting?
- Write the unknown parameter vector for the sphere linearization.
- How is the sphere fit solved according to the notes?

Correct / accepted:

- Row i : $[1, -2x_i, -2y_i, -2z_i]$.
- Parameter vector: $[x_0^2 + y_0^2 + z_0^2 - R^2, x_0, y_0, z_0]^T$.
- Right side: $-(x_i^2 + y_i^2 + z_i^2)$; solve using the pseudo-inverse.

Likely traps / distractors:

- Use only three columns as in the 2D circle.
- Right side $-R^2$.
- Use a homogeneous zero right-hand side.

Why it is correct. The sphere expansion introduces exactly one more center coordinate and one more squared data term than the circle.

Memory / understanding hook. Take the circle matrix and append $-2z_i$; append z_0 to the unknowns.

Q21. General implicit ellipse equation – Priority: VERY HIGH*Source anchor: Circle/Ellipse notes, Sec. 1.2***Possible tricky formulations**

- Write the six-parameter quadratic form used for ellipse fitting.
- Which terms appear in the general implicit conic equation in the notes?
- What is the coefficient vector order used later in $\mathbf{z}$?

Correct / accepted:

- $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$.

- Parameter order: $\mathbf{z} = [A \ B \ C \ D \ E \ F]^T$.

Likely traps / distractors:

- Omit the cross term Bxy .
- Use $Dx^2 + Ey^2$ instead of linear $Dx + Ey$.
- Use only five coefficients.

Why it is correct. The notes use the general second-degree implicit curve with six coefficients and then stack those coefficients into $\mathbf{z}$.

Memory / understanding hook. Quadratic terms first (x^2, xy, y^2) , then linear terms (x, y) , then constant.

Q22. Condition for the conic to be an ellipse – Priority: VERY HIGH

Source anchor: Circle/Ellipse notes, Sec. 1.2

Possible tricky formulations

- What inequality distinguishes the ellipse case in these notes?
- For $Ax^2 + Bxy + Cy^2 + \dots = 0$, which discriminant condition is required?
- Which sign of $B^2 - 4AC$ is associated with an ellipse here?

Correct / accepted:

- $B^2 - 4AC < 0$.

Likely traps / distractors:

- $B^2 - 4AC > 0$.
- $B^2 - 4AC = 0$.
- $A^2 - 4BC < 0$.

Why it is correct. The source explicitly states that the quadratic represents an ellipse when $B^2 - 4AC < 0$.

Memory / understanding hook. Ellipse means the conic discriminant is negative: $B^2 - 4AC < 0$.

Q23. Ellipse fitting matrix M and homogeneous equation – Priority: VERY HIGH

Source anchor: Circle/Ellipse notes, Sec. 1.2

Possible tricky formulations

- How is the ellipse equation written for N data points?
- What is one row of M ?
- What equation relates M and the coefficient vector $\mathbf{z}$?

Correct / accepted:

- $M\mathbf{z} = \mathbf{0}$.
- Row i of M : $[x_i^2, x_i y_i, y_i^2, x_i, y_i, 1]$.
- $\mathbf{z} = [A, B, C, D, E, F]^T$.

Likely traps / distractors:

- Use row $[x_i, y_i, 1]$ (plane-like, not conic).
- Use $M\mathbf{z} = \mathbf{1}$.
- Swap the right side to measured y_i .

Why it is correct. Substituting each point into the implicit quadratic gives a linear equation in the six coefficients with zero on the right.

Memory / understanding hook. Ellipse row mirrors the six terms of the conic equation, in exactly the same order.

Q24. Why ellipse parameters need a scale constraint – Priority: VERY HIGH*Source anchor: Circle/Ellipse notes, Sec. 1.2***Possible tricky formulations**

- Why is $M\mathbf{z} = 0$ not enough to determine a unique nonzero $\mathbf{z}$?
- What ambiguity exists in homogeneous ellipse parameters?
- Why can the discriminant be fixed to a particular negative value?

Correct / accepted:

- The scale of $\mathbf{z}$ is arbitrary: if $\mathbf{z}$ is a solution, any nonzero scalar multiple represents the same implicit curve.
- The notes fix the scale using $B^2 - 4AC = -1$.

Likely traps / distractors:

- The center is arbitrary.
- The data points are arbitrary, so no constraint is possible.
- Set $A + B + C + D + E + F = 0$ as the source constraint.

Why it is correct. A homogeneous implicit equation is unchanged by multiplying all coefficients by the same nonzero constant, so a normalization removes that length/scale ambiguity.

Memory / understanding hook. Homogeneous coefficients have no natural size; force one scalar condition to pin the size down.

Q25. Matrix form of the ellipse constraint – Priority: VERY HIGH*Source anchor: Circle/Ellipse notes, Sec. 1.2***Possible tricky formulations**

- How is $B^2 - 4AC = -1$ written as a quadratic form?
- What is the role of matrix K ?
- Choose the correct compact normalization equation.

Correct / accepted:

- $\mathbf{z}^T K \mathbf{z} = -1$.

$$\bullet K = \begin{bmatrix} 0 & 0 & -2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ -2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Likely traps / distractors:

- $\mathbf{z}^T K \mathbf{z} = 0$.
- $K = I$.
- $\mathbf{z}^T \mathbf{z} = -1$.

Why it is correct. The nonzero entries of K generate $B^2 - 4AC$ when the quadratic form is expanded.

Memory / understanding hook. K only touches A, B, C : the ellipse test depends only on the quadratic terms.

Q26. Ellipse fitting optimization problem – Priority: VERY HIGH*Source anchor: Circle/Ellipse notes, Sec. 1.2***Possible tricky formulations**

- State the constrained minimization used for ellipse fitting.
- What quantity is minimized subject to the ellipse normalization?
- Why not simply solve $M\mathbf{z} = 0$ exactly for noisy data?

Correct / accepted:

$$\bullet \quad \boxed{\mathbf{z} = \arg \min_{\mathbf{z}} \|\mathbf{M}\mathbf{z}\| \quad \text{subject to } \mathbf{z}^T \mathbf{K} \mathbf{z} = -1}.$$

- Equivalent squared objective used in the Lagrangian: minimize $\mathbf{z}^T \mathbf{M}^T \mathbf{M} \mathbf{z}$ under the same constraint.

Likely traps / distractors:

- Minimize $\|\mathbf{z}\|$ with no data term.
- Maximize $\|\mathbf{M}\mathbf{z}\|$.
- Use $\mathbf{M}\mathbf{z} = \mathbf{1}$ and no constraint.

Why it is correct. The residual of the homogeneous equations is $\mathbf{M}\mathbf{z}$, and the constraint both enforces the chosen ellipse normalization and prevents the trivial zero vector.

Memory / understanding hook. Ellipse fit = make $\mathbf{M}\mathbf{z}$ small, but forbid the trivial scale by $\mathbf{z}^T \mathbf{K} \mathbf{z} = -1$.

Q27. Lagrangian for ellipse fitting – Priority: VERY HIGH

Source anchor: Circle/Ellipse notes, Sec. 1.2

Possible tricky formulations

- Write the Lagrangian used to enforce the ellipse constraint.
- Which term is multiplied by the Lagrange multiplier λ ?
- How is $\|\mathbf{M}\mathbf{z}\|^2$ written as a quadratic form?

Correct / accepted:

$$\bullet \quad \boxed{L(\mathbf{z}, \lambda) = \mathbf{z}^T \mathbf{M}^T \mathbf{M} \mathbf{z} + \lambda(\mathbf{z}^T \mathbf{K} \mathbf{z} + 1)}.$$

Likely traps / distractors:

- $\mathbf{z}^T \mathbf{M} \mathbf{z} + \lambda(\mathbf{z}^T \mathbf{z} - 1)$.
- $\|\mathbf{M}\mathbf{z}\| + \lambda \mathbf{z}$ as a vector equation.
- Use -1 inside the parentheses: $\mathbf{z}^T \mathbf{K} \mathbf{z} - 1$ (does not encode the source constraint $= -1$).

Why it is correct. The squared residual is $\mathbf{z}^T \mathbf{M}^T \mathbf{M} \mathbf{z}$, and $\mathbf{z}^T \mathbf{K} \mathbf{z} + 1 = 0$ is the constraint in zero-equality form.

Memory / understanding hook. Objective plus $\lambda \times (\text{constraint moved to zero})$.

Q28. Gradient condition for ellipse fitting – Priority: VERY HIGH

Source anchor: Circle/Ellipse notes, Sec. 1.2

Possible tricky formulations

- Differentiate the ellipse Lagrangian with respect to $\mathbf{z}$.
- What stationarity equation is obtained before the generalized eigenproblem?
- Which factor of 2 appears in the gradient?

Correct / accepted:

$$\bullet \quad \boxed{2\mathbf{M}^T \mathbf{M} \mathbf{z} + 2\lambda \mathbf{K} \mathbf{z} = \mathbf{0}}.$$

- $\mathbf{M}^T \mathbf{M} \mathbf{z} = -\lambda \mathbf{K} \mathbf{z}$ after dividing by 2 and rearranging.

Likely traps / distractors:

- $\mathbf{M}\mathbf{z} + \lambda \mathbf{K} = \mathbf{0}$.
- $2\mathbf{M}\mathbf{z} + 2\lambda \mathbf{z} = \mathbf{0}$.
- $\mathbf{M}^T \mathbf{M} \mathbf{z} = \lambda \mathbf{z}$ with $\mathbf{K}$ omitted.

Why it is correct. Both $\mathbf{M}^T \mathbf{M}$ and $\mathbf{K}$ are used inside quadratic forms, so the symmetric-matrix derivative rule produces a factor 2 for each term.

Memory / understanding hook. Differentiate quadratic forms: $2(\text{matrix})(\text{vector})$; then move the constraint term to the other side.

Q29. Generalized eigenvalue problem for ellipse fitting – Priority: VERY HIGH*Source anchor: Circle/Ellipse notes, Sec. 1.2***Possible tricky formulations**

- What generalized eigenvalue equation appears in ellipse fitting?
- Which method replaces the pseudo-inverse for the constrained ellipse problem?
- After applying the Lagrangian, what equation produces candidate coefficient vectors?

Correct / accepted:

- $M^T M \mathbf{z} = -\lambda K \mathbf{z}$.
- This is the **generalized eigenvalue problem** identified in the notes.

Likely traps / distractors:

- $M \mathbf{z} = \lambda \mathbf{z}$.
- $K \mathbf{z} = 0$.
- $A^T A \mathbf{x} = A^T \mathbf{b}$ as the final ellipse method.

Why it is correct. The stationarity condition has one matrix multiplying $\mathbf{z}$ on each side, with $-\lambda$ scaling the $K \mathbf{z}$ side.

Memory / understanding hook. Ellipse fitting is the one that ends in “matrix vs. matrix eigenvalue form, not a simple pseudo-inverse.

Q30. Selecting among ellipse candidates – Priority: HIGH*Source anchor: Circle/Ellipse notes, Sec. 1.2***Possible tricky formulations**

- How many candidate $\mathbf{z}$ solutions can be obtained according to the notes?
- How is the optimal candidate selected?
- What objective value is compared among generalized eigenvector candidates?

Correct / accepted:

- At most **six** candidate solutions for $\mathbf{z}$ are obtained.
- Choose the candidate minimizing $\mathbf{z}^T M^T M \mathbf{z}$ while satisfying the normalization.

Likely traps / distractors:

- Exactly two candidates in all cases.
- Choose the largest residual.
- Choose based only on $\|\mathbf{z}\|$ without the fit residual.

Why it is correct. The notes explicitly state “at most six candidates and select the one with the smallest squared residual objective.

Memory / understanding hook. Six coefficients can yield up to six generalized-eigen candidates; keep the one that fits the data best.

Q31. What is a curve normal? – Priority: VERY HIGH*Source anchor: Circle/Ellipse notes, Sec. 2***Possible tricky formulations**

- How are curve normals obtained from an implicit function $J(x, y)$?
- Which vector is perpendicular to a level curve $J = 0$?
- Complete: the curve normal is given by the gradient of J .

Correct / accepted:

- $\nabla J = [\partial J / \partial x, \partial J / \partial y]^T$.
- The normal direction is the gradient of the implicit curve function.

Likely traps / distractors:

- The tangent vector itself.
- The Hessian.
- The parameter vector $\mathbf{z}$ regardless of the point.

Why it is correct. The curve-normal section states directly that normals are given by gradients.

Memory / understanding hook. Level sets are perpendicular to the direction of fastest increase: the gradient.

Q32. Normal of a circle – Priority: VERY HIGH

Source anchor: Circle/Ellipse notes, Sec. 2

Possible tricky formulations

- Find the normal vector to $J = (x - x_0)^2 + (y - y_0)^2 - R^2$.
- What is the gradient of the implicit circle function?
- Which vector points radially from the center (up to scale)?

Correct / accepted:

- $\nabla J = \begin{bmatrix} 2(x - x_0) \\ 2(y - y_0) \end{bmatrix}$.
- Any nonzero scalar multiple, such as $[x - x_0, y - y_0]^T$, has the same normal direction.

Likely traps / distractors:

- $[2x_0, 2y_0]^T$.
- $[y - y_0, -(x - x_0)]^T$ (a tangent-like perpendicular direction).
- $[R, R]^T$.

Why it is correct. Differentiate the two squared coordinate differences; the constant $-R^2$ disappears.

Memory / understanding hook. Circle normal points from the center to the point; the factor 2 does not change direction.

Q33. Normal of an ellipse / general conic – Priority: VERY HIGH

Source anchor: Circle/Ellipse notes, Sec. 2

Possible tricky formulations

- Differentiate $J = Ax^2 + Bxy + Cy^2 + Dx + Ey + F$ with respect to x and y .
- Which gradient gives the normal to the fitted ellipse?
- What are the two components of the conic gradient?

Correct / accepted:

- From direct differentiation of the printed J : $\nabla J = \begin{bmatrix} 2Ax + By + D \\ Bx + 2Cy + E \end{bmatrix}$.
- Source caveat: the final line of the PDF prints the second component as $2Cx + Bx + E$; this is inconsistent with differentiating Cy^2 and is best treated as a typographical error.

Likely traps / distractors:

- $[2Ax + By + D, 2Cx + Bx + E]^T$ as a mathematically consistent derivative (it matches the source print but not the displayed J).
- $[Ax + By + D, Cx + Bx + E]^T$.
- $[A, B, C, D, E, F]^T$.

Why it is correct. For the y derivative, $Bxy \rightarrow Bx$ and $Cy^2 \rightarrow 2Cy$; the x derivative is $2Ax + By + D$.

Memory / understanding hook. Differentiate term-by-term: in the second component, the y^2 term must leave a y , not an x .

Q34. Circle vs. ellipse: inhomogeneous vs. homogeneous – Priority: HIGH

Source anchor: Both notes; Circle/Ellipse Secs. 1.1-1.2

Possible tricky formulations

- Which fitting problem has a zero right-hand side?
- Which problem is described as inhomogeneous?
- Match method to equation: circle vs. ellipse.

Correct / accepted:

- Circle fitting: **inhomogeneous** linear system with nonzero right-hand side.
- Ellipse fitting: **homogeneous** system $M\mathbf{z} = 0$ plus a normalization constraint.

Likely traps / distractors:

- Both are homogeneous.
- Both use exactly the same pseudo-inverse equation.
- Ellipse has a measured y vector on the right like line fitting.

Why it is correct. The circle rearrangement puts $-(x_i^2 + y_i^2)$ on the right, while every ellipse point satisfies a zero-valued implicit equation.

Memory / understanding hook. Circle: data on both sides. Ellipse: all conic terms stay on the left and equal zero.

Q35. Which fitting method uses which solver? – Priority: HIGH

Source anchor: Circle/Ellipse notes

Possible tricky formulations

- Match circle, sphere, and ellipse fitting to pseudo-inverse or generalized eigenproblem.
- Which fit ends with $M^T M\mathbf{z} = -\lambda K\mathbf{z}$?
- Which fits are solved by pseudo-inverse according to the source?

Correct / accepted:

- Circle: inverse/pseudo-inverse.
- Sphere: pseudo-inverse.
- Ellipse: constrained minimization $\rightarrow$ generalized eigenvalue problem.

Likely traps / distractors:

- Ellipse: simple inverse of M .
- Circle: generalized eigenproblem with K .
- Sphere: no linear solution.

Why it is correct. The notes explicitly separate the inhomogeneous circle/sphere linear solves from the constrained homogeneous ellipse solve.

Memory / understanding hook. Round shapes with center/radius use pseudo-inverse here; six-coefficient ellipse uses the K -constrained eigenproblem.

Q36. Matrix dimensions and parameter counts – Priority: MEDIUM-HIGH

Source anchor: Directly read from the source systems

Possible tricky formulations

- How many unknown parameters appear in each linearized fitting system?
- What is the number of columns in the circle, sphere, line, plane, and ellipse design matrices?
- Identify a system from its row length.

Correct / accepted:

- Line: 2 columns/unknowns (m, b) . Plane: 3 (a, b, c) . Circle linearization: 3 grouped unknowns. Sphere: 4

grouped unknowns. Ellipse: 6 coefficients (A, B, C, D, E, F).

Likely traps / distractors:

- Circle uses only 2 unknowns because the center is 2D (radius/grouped constant adds another).
- Ellipse uses 5 because F can always be dropped (not in the source formulation).
- Plane uses 4 because z is an unknown parameter (it is measured data here).

Why it is correct. The column count is visible directly in each matrix equation and equals the length of the parameter vector.

Memory / understanding hook. 2 line, 3 plane/circle, 4 sphere, 6 ellipse.

Q37. Minimum-point statements actually supported by the notes – Priority: MEDIUM

Source anchor: Matrix notes, Sec. 4.1; other counts are not explicitly stated

Possible tricky formulations

- How many points do the notes explicitly say are needed to estimate a 2D line?
- Do the notes explicitly state the minimum number of circle/ellipse points?
- Which minimum-data claim is safe to quote verbatim from the source?

Correct / accepted:

- The notes explicitly say: **if at least two points are given, the line can be estimated.**
- The provided PDFs do **not explicitly state** minimum point counts for circle, sphere, plane, or ellipse fitting.

Likely traps / distractors:

- Claim the PDFs explicitly say “three points for a circle (they do not).”
- Claim the PDFs explicitly say “five points for an ellipse (they do not).”

Why it is correct. Only the line-fitting section contains an explicit minimum-point sentence. Other minimum counts may be derivable from parameter counts, but they are not stated in the supplied notes.

Memory / understanding hook. For source-faithful quiz answers: memorize “line: at least 2; treat other minima as outside the written notes unless the lecturer taught them elsewhere.”

Q38. Recovering the circle radius from the fitted grouped parameter (derived) – Priority: MEDIUM

Source anchor: Derived directly from Circle/Ellipse Sec. 1.1 equation

Possible tricky formulations

- If the first fitted circle parameter is $q = x_0^2 + y_0^2 - R^2$, how can R be recovered?
- After solving the linearized circle system, what relation links q , the center, and the radius?

Correct / accepted:

- From $q = x_0^2 + y_0^2 - R^2$, $R = \sqrt{x_0^2 + y_0^2 - q}$ (taking the nonnegative radius).

Likely traps / distractors:

- $R = q - x_0^2 - y_0^2$.
- $R = \sqrt{q - x_0^2 - y_0^2}$.
- $R = x_0 + y_0 - q$.

Why it is correct. This formula is not written as a separate line in the PDF, but it follows algebraically from the first component of the parameter vector shown there.

Memory / understanding hook. Move q to the other side: $R^2 = \text{center-squared sum} - q$.

Q39. Source typo traps you should not memorize – Priority: HIGH AS A TRAP

Source anchor: Both PDFs

Possible tricky formulations

- Which displayed lines in the notes appear internally inconsistent?
- If a quiz reproduces a source typo, how can you identify the intended formula from neighboring equations?
- Spot the typo: derivative proof, least-squares expansion, sphere formula, ellipse normal.

Correct / accepted:

- Matrix p.2 proof line prints $\partial J / \partial x_i = 2x_i^2$, while the displayed result and basic differentiation give $2x_i$.
- Matrix p.4 expansion visually contains a cross-term inconsistent with the next gradient; the intended least-squares cross-term is $-2\mathbf{x}^T A^T \mathbf{b}$.
- Circle/Ellipse p.1 prints $(z - y_0)^2$ in the sphere formula, while p.2 expansion uses z_0 ; intended is $(z - z_0)^2$.
- Circle/Ellipse p.4 prints $2Cx + Bx + E$ for the second ellipse-gradient component; differentiating Cy^2 gives $2Cy$, so the consistent component is $Bx + 2Cy + E$.

Likely traps / distractors:

- Memorize every printed typo literally without checking consistency.

Why it is correct. Each issue can be detected by comparing the line to the equation immediately before or after it, without importing an unrelated method.

Memory / understanding hook. When a symbol looks suspicious, differentiate/expand one line yourself and compare with the next displayed result.

Q40. One-step method recognition (mixed rapid-fire) – Priority: MEDIUM-HIGH

Source anchor: Both PDFs

Possible tricky formulations

- Identify the method from the formula: $(A^T A)^{-1} A^T b$, $M^T M z = -\lambda K z$, $[1, -2x_i, -2y_i]$, or $[x_i, y_i, 1]$.
- Which concept is being tested if you see $B^2 - 4AC$?
- Which concept is being tested if you see ∇J ?

Correct / accepted:

- $(A^T A)^{-1} A^T b$: least squares / normal equation solution.
- $M^T M z = -\lambda K z$: ellipse generalized eigenproblem.
- $[1, -2x_i, -2y_i]$: circle fitting row.
- $[x_i, y_i, 1]$: plane fitting row.
- $B^2 - 4AC$: conic/ellipse condition. ∇J : curve normal.

Likely traps / distractors:

- Treat every matrix formula as the same fitting problem.
- Confuse $[x_i, 1]$ (line) with $[x_i, y_i, 1]$ (plane).

Why it is correct. The notes repeatedly use distinctive row patterns and terminal equations for each method; recognizing those signatures is faster than re-deriving everything under time pressure.

Memory / understanding hook. Memorize the signature, not just the chapter: line $[x, 1]$, plane $[x, y, 1]$, circle $[1, -2x, -2y]$, ellipse $[x^2, xy, y^2, x, y, 1]$.

30-second final recall checklist

- $\nabla(\mathbf{x}^T \mathbf{x}) = 2\mathbf{x}$.
- $\nabla(\mathbf{x}^T \mathbf{a}) = \mathbf{a}$.
- $\nabla(\mathbf{x}^T A \mathbf{x}) = (A^T + A)\mathbf{x}$.
- Symmetric B : $2B\mathbf{x}$.
- Over-determined: $m > n$.
- Least squares: minimize $\|A\mathbf{x} - \mathbf{b}\|_2^2$.
- Normal equation: $A^T A \mathbf{x} = A^T \mathbf{b}$.
- Line row: $[x, 1]$.
- Plane row: $[x, y, 1]$.
- Circle row: $[1, -2x, -2y]$.
- Sphere row adds $-2z$.
- Ellipse row: $[x^2, xy, y^2, x, y, 1]$.
- Ellipse condition: $B^2 - 4AC < 0$.
- Scale fix: $B^2 - 4AC = -1$.
- Compact constraint: $\mathbf{z}^T K \mathbf{z} = -1$.
- Generalized eigenproblem: $M^T M \mathbf{z} = -\lambda K \mathbf{z}$.
- Circle normal: radial gradient.
- Ellipse normal: differentiate the conic term-by-term.

Source map and caveats

This guide is based only on the two supplied PDFs: *Derivation of scalar functions with respect to vectors* (5 pages) and *Circle and Ellipse Fitting* (4 pages). The first supplies notation, vector derivatives, least squares, line fitting, and plane fitting. The second supplies circle/sphere fitting, ellipse fitting with a constrained generalized eigenvalue problem, and curve normals.

Four source-level typo/inconsistency alerts worth remembering:

1. In the $\mathbf{x}^T \mathbf{x}$ proof, one component derivative is printed as $2x_i^2$, but the displayed final result is $2\mathbf{x}$ and termwise differentiation gives $2x_i$.
2. In the least-squares expansion, the displayed cross-term is visually inconsistent with the next gradient; the intended residual expansion uses $-2\mathbf{x}^T A^T \mathbf{b}$.
3. The first sphere equation prints $(z - y_0)^2$, but its next-page expansion and the stated center $[x_0, y_0, z_0]^T$ imply $(z - z_0)^2$.
4. The printed ellipse normal has $2Cx + Bx + E$ in the second component; direct differentiation of Cy^2 gives $2Cy$, hence $Bx + 2Cy + E$.

Last-minute strategy

If you have very little time, master Questions 1–7, 11–18, 21–29, and 31–35 first. Those cover nearly every major derivation and method transition in the supplied notes. Then use the final checklist until you can reproduce each formula without looking.